

A universal scheme to self-test any quantum state or measurement

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The emergence of quantum devices has raised a significant issue: how to certify the quantum properties of a device without placing trust in it? To characterize quantum states and measurements in a device-independent way, up to some degree of freedom, we can make use of a technique known as self-testing. Although schemes have been proposed to self-test all pure multipartite entangled states (up to complex conjugation) and real local projective measurements, little has been done to certify mixed entangled states, composite or non-projective measurements. By using the framework of quantum networks, we propose a scheme for self-testing (up to complex conjugation) arbitrary extremal measurements, including the projective ones. This then allows us to propose an indirect way to self-test any quantum state, including mixed ones, as well as any quantum measurement, including non-extremal ones. The quantum network considered here is the simple star network, which is implementable using current technologies. For our purposes, we also construct a scheme that can be used to self-test the two-dimensional tomographically complete set of measurements with an arbitrary number of parties.

Demonstrating the strength of quantum systems over classical ones in tasks like computation and cryptography has been one of the most important achievements in quantum information theory. Quantum protocols often offer greater efficiency and security than their classical counterparts. However, to implement them, it is necessary to certify the quantumness of the devices involved. The traditional methods of certifying a quantum state or a measurement rely on trusting the devices used and are, therefore, referred to as device-dependent schemes. For instance, any quantum state can be device-dependently verified through, for example, the quantum tomography if one has complete knowledge of the measurement apparatuses. However, from a practical perspective, it is unreasonable to completely trust the device. Therefore, it is desirable to devise certification schemes that require few or even minimal assumptions about the devices used such as, for instance, the device-independent (DI) ones. These schemes enable the verification of the non-trivial quantum properties of an unknown device solely through the statistical data it produces. The essential

resource underlying any DI scheme is Bell non-locality—the existence of quantum correlations that cannot be reproduced by any local (or simply classical) hidden-variable model^{1–3}. In particular, observing a violation of a Bell inequality certifies, in a fully DI manner, that the device operates on an entangled state.

Interestingly, the observation of Bell non-locality in the data produced by a given device can provide a lot more information about its internal working than just the presence of entanglement. The strongest and the most complete form of Bell non-locality-based certification is self-testing^{4,5}. It enables almost a full (up to certain well-understood equivalences) characterization of the underlying quantum state and the measurements performed on it.

In recent years, a plethora of schemes have been proposed to self-test pure quantum states or quantum measurements^{6–20}. In particular, refs. 13,19 propose self-testing strategies for any pure entangled bipartite or multipartite states, respectively, where the second method is based on the quantum networks scenario. The only scheme

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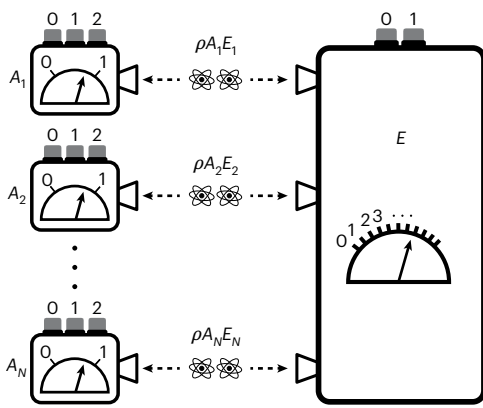


Fig. 1 | Depiction of the quantum network scenario. It consists of $N + 1$ parties, namely, A_i ($i = 1 \dots N$), and E and N independent sources distributing bipartite quantum states $\rho_{A_i E_i}$ among the parties. The central party E shares quantum states with each one of the other external parties A_i . Although each A_i has three inputs and two outcomes, E has two inputs. The first Eve's measurement has $2^N - 1$ outcomes, whereas the second one has K outcomes.

to self-test a mixed entangled state, particularly a bound entangled state, was only recently proposed in ref. 20 (see also refs. 21, 22). As for quantum measurements, apart from a bunch of schemes allowing to certify various examples of quantum measurements^{14, 16, 17, 23, 24}, including the composite ones^{20, 25–27}, a general method to certify any real local projective measurement has only very recently been designed²⁴. Despite this progress, there exists no single unified scheme that allows for the certification of quantum measurements and quantum states (pure or mixed). Moreover, no general scheme for the certification of composite measurements based on quantum networks has been proposed.

The main goal of this work is to tackle these challenges by providing a universal scheme that can self-test (up to complex conjugation) any extremal generalized quantum measurement (positive operator-valued measure (POVM) on any finite-dimensional Hilbert space. Since all the projective measurements are extremal, they are included, too. We then extend the scheme to indirectly self-test any non-extremal quantum measurement. On the other hand, our method enables the DI-certified remote preparation of quantum states, which can also be seen as an indirect way of DI certifying any quantum state, even mixed ones. Overall, our approach offers a unified way to self-test both arbitrary quantum states and measurements.

We utilize the framework of quantum networks consisting of an arbitrary number of parties and sources (as depicted in Fig. 1 and explained later in the text). We first propose a Bell inequality that can be used to self-test the two-dimensional tomographically complete set of measurements with arbitrary number of parties. On the basis of this, we are then able to certify any extremal POVM and eventually any quantum state and quantum measurement. It is worth mentioning that unlike the standard self-testing statements, our scheme requires additional causality constraints that are natural in quantum networks, particularly that the sources are statistically independent as that in ref. 19. Our scheme is simple and can be implemented using the current infrastructure of quantum networks^{28–30}.

Before proceeding to the main results, let us first introduce the scenario and relevant notions required throughout this work.

Preliminaries

The quantum network scenario under scrutiny is composed of an external group of N Alices, denoted as A_i ($i = 1 \dots N$), and a central party E called Eve. All the $N + 1$ parties are spatially separated. Correspondingly, N independent sources distribute bipartite quantum states—generally entangled—among the parties (Fig. 1). These states are denoted by $\rho_{A_i E_i}$ where the subsystems A_i will be measured by the external parties,

whereas the other subsystems E_i go to Eve. Within quantum theory, the joint state of statistically independent sources is represented as a product of the individual states, that is,

$$\rho_{AE} = \bigotimes_{i=1}^N \rho_{A_i E_i}, \quad (1)$$

where we denoted $A := A_1 \dots A_N$ and $E := E_1 \dots E_N$. On their shares of the joint state ρ_{AE} , each external party A_i performs one of three available measurements, each with two outcomes; the measurement choices and outcomes of party A_i are denoted $x_i = 0, 1, 2$ and $a_i = 0, 1$, respectively. At the same time, the central party Eve freely chooses to perform one of two measurements, where the first one has 2^N outcomes, whereas the second one has $K \leq 2^N$ outcomes. We denote Eve's measurement choices by $e = 0, 1$ and outcomes by $l = 0 \dots 2^N - 1$ for $e = 0$ and $l = 0 \dots K - 1$ for $e = 1$. Let us add here that the first Eve's measurement is used to certify the external parties' measurements as well as the states prepared by the sources, whereas the second measurement is the one to be certified. Importantly, the parties cannot communicate classically during the experiment.

After running many rounds of the above measurements, all the parties can bring their data together to reconstruct the set of probability distributions $\vec{p} = \{p(\mathbf{a}|\mathbf{x}\mathbf{e})\}$, which describe the correlations that are produced between all the observers, where each $p(\mathbf{a}|\mathbf{x}\mathbf{e})$ is the probability of obtaining outcomes $a_1 \dots a_N = \mathbf{a}$ by all the Alices and l by Eve after they perform the measurements labelled by $x_1 \dots x_N = \mathbf{x}$ and e , respectively. From Born's rule, we can write

$$p(\mathbf{a}|\mathbf{x}\mathbf{e}) = \text{Tr} \left[\rho_{AE} \left(\bigotimes_{i=1}^N M_{a_i|x_i} \otimes R_{l|e} \right) \right], \quad (2)$$

where $M_{i,x_i} = \{M_{a_i|x_i}\}$, for every i , and $E_e = \{R_{l|e}\}$ are the measurements performed by parties A_i and E . The measurement elements $M_{a_i|x_i}$ and $R_{l|e}$ are positive semidefinite and sum up to the identity on the respective Hilbert space for every measurement choice x_i or e of all $N + 1$ parties.

To better understand the observed correlations, expressing them in terms of the expected values of external observables can be helpful. These values are defined as

$$\langle A_{1,x_1} \dots A_{N,x_N} R_{l|e} \rangle = \sum_{a_1 \dots a_N=0}^1 (-1)^{\sum_{i=1}^N a_i} p(\mathbf{a}|\mathbf{x}\mathbf{e}). \quad (3)$$

It is important to note that by using equation (2), these expectation values can be expressed as $\langle A_{1,x_1} \dots A_{N,x_N} R_{l|e} \rangle = \text{Tr}[(\bigotimes_{i=1}^N A_{i,x_i}) \otimes R_{l|e} \rho_{AE}]$, where A_{i,x_i} are quantum operators defined via the measurement elements as $A_{i,x_i} = M_{0|x_i} - M_{1|x_i}$ for every x_i and i . Notice that in a particular case when the measurement $\{M_{0|x_i}, M_{1|x_i}\}$ is projective, the corresponding operator A_{i,x_i} becomes a standard Hermitian observable with ± 1 eigenvalues. For our convenience, in the above representation, Eve's measurements remain represented in terms of the measurement elements $R_{l|e}$ instead of observables.

Let us now introduce the idea of self-testing by referring back to the scenario shown in Fig. 1. First, we assume that the measurements conducted by the parties and the states $\rho_{A_i E_i} \in \mathcal{L}(\mathcal{H}_{A_i} \otimes \mathcal{H}_{E_i})$ for $i = 1 \dots N$ are unknown. The only knowledge that Alices and Eve have about the whole system is encoded in the observed correlations $\vec{p}$. It is worth pointing out already now that since the dimensions of the local Hilbert spaces $\mathcal{H}_{A_i}$ and $\mathcal{H}_{E_i}$ are unspecified, we can utilize the standard dilation argument and presume that the shared states are pure, that is, $\rho_{A_i E_i} = |\psi_{A_i E_i}\rangle\langle\psi_{A_i E_i}|$. On the same ground, we can also assume that the measurements of the external parties are projective. However, this assumption can be dropped by using a more general sum of squares decomposition of the Bell operators corresponding to the inequalities (equation (7)) that take into account that the measurements are not projective.

Let us then consider another reference experiment giving rise to the same correlations $\vec{p}$, which is performed on known quantum states shared by the parties $|\psi_{A_i E_i}\rangle \in \mathcal{H}_{A_i'} \otimes \mathcal{H}_{E_i'}$ and with known measurements A_{i,x_i}' and E_{i,y_i}' of Alices and Eve, respectively, where $\mathcal{H}_{A_i'}$ and $\mathcal{H}_{E_i'}$ are some Hilbert spaces of a known dimension. The task of self-testing is to deduce from the observed $\vec{p}$ that the actual experiment is equivalent to the reference one in the following sense: (1) the local Hilbert spaces admit the product form $\mathcal{H}_{A_i} = \mathcal{H}_{A_i'} \otimes \mathcal{H}_{A_i''}$ and $\mathcal{H}_{E_i} = \mathcal{H}_{E_i'} \otimes \mathcal{H}_{E_i''}$ for some auxiliary Hilbert spaces $\mathcal{H}_{A_i''}$ and $\mathcal{H}_{E_i''}$; and (2) there are local unitary operations $U_{A_i} : \mathcal{H}_{A_i} \rightarrow \mathcal{H}_{A_i'} \otimes \mathcal{H}_{A_i''}$ and $U_{E_i} : \mathcal{H}_{E_i} \rightarrow \mathcal{H}_{E_i'} \otimes \mathcal{H}_{E_i''}$ such that

$$(U_{A_i} \otimes U_{E_i}) |\psi_{A_i E_i}\rangle = |\psi_{A_i' E_i'}\rangle \otimes |\xi_{A_i'' E_i''}\rangle, \quad (4)$$

where $|\xi_{A_i'' E_i''}\rangle \in \mathcal{H}_{A_i''} \otimes \mathcal{H}_{E_i''}$ is some auxiliary quantum state, and

$$U_{A_i} A_{i,x_i} U_{A_i}^\dagger = A_{i,x_i}' \otimes \mathbb{1}_{A_i''}, \quad U_{E_i} R_{i,y_i} U_{E_i}^\dagger = R_{i,y_i}' \otimes \mathbb{1}_{E_i''}, \quad (5)$$

where $\mathbb{1}_{E_i''}$ is the identity acting on the Eve's auxiliary system $\mathcal{H}_{E_i''}$, and $U_E = \bigotimes_{i=1}^N U_{E_i}$. We use the notation $E'' = E_1'' \dots E_N''$.

If the above conditions (1) and (2) are met, one says that the reference state and measurements are self-tested in the actual experiment from the observed correlations. Moreover, for any outcome l of Eve's measurement, the post-measurement states $\rho_A^{l|e}$ held by the external parties A_i satisfy

$$U_A \rho_A^{l|e} U_A^\dagger = \tilde{\rho}_A^{l|e} \otimes \xi_{A''}^{l|e}, \quad (6)$$

where $\tilde{\rho}_A^{l|e}$ are the reference post-measured states, $\xi_{A''}^{l|e}$ are some auxiliary states and $U_A = U_{A_1} \otimes \dots \otimes U_{A_N}$, where U_{A_i} are the same unitary transformations as those in equations (4) and (5), then we say that the post-measurement states are self-tested, too. We refer to this method of self-testing as the DI-certified remote preparation of quantum states. Unlike standard self-testing, this approach allows self-testing probabilistically as the state to be certified occurs with the probability of obtaining the outcome l by Eve as $p(l|e)$. It should be noted here that the above definition of self-testing the sources (equation (4)), measurements (equation (5)) and the post-measurement states (equation (6)) holds up to the complex conjugation of the reference states and measurements. The reason is that the correlations $\vec{p}$ are invariant under the complex conjugation of quantum states and measurements.

Let us finally mention that throughout this work, we assume that the local states of $|\psi_{A_i E_i}\rangle$ are full rank as the local measurements can only be certified on the supports of the local density matrices.

Results

We are now ready to present our scheme for the self-testing of composite measurements and quantum states. It is divided into three major parts. The first one involves the self-testing of the two-dimensional tomographically complete set of Pauli measurements in the measurements of the external parties and the two-qubit Bell states in the states generated by all the sources. The second one involves self-testing any quantum measurement performed by the central party using the states and external parties' measurements as certified in the above first part of the scheme. Last, the third part is concerned with applying the first two parts for self-testing the post-measurement state at the external parties' laboratories. As a result, our schemes enable self-testing any quantum state (pure or mixed) distributed between the external parties by Eve's measurements, as well as facilitating the self-testing of quantum measurements performed by the central party.

Part 1

To self-test the tomographically complete set of measurements in the external parties' measurement devices and the two-qubit singlets in

the states generated by the sources, we introduce the following class of 2^N Bell inequalities that are suitable modifications of those introduced in refs. 20,31:

$$I_l = (-1)^{l_1} \left[(N-1) \left\langle \tilde{A}_{1,1} \prod_{i=2}^N A_{i,1} \right\rangle + \sum_{i=2}^N (-1)^{l_i} \left\langle \tilde{A}_{1,0} A_{i,0} \right\rangle - (-1)^{l_1} \sum_{i=2}^N (-1)^{l_i} \left\langle A_{1,2} A_{i,2} \prod_{\substack{j=2 \\ j \neq i}}^N A_{j,1} \right\rangle \right] \leq \beta_C, \quad (7)$$

where $l \equiv l_1 \dots l_N$ such that $l_1, l_2, \dots, l_N = 0, 1$ is the binary representation of the outcome l and

$$\tilde{A}_{1,0} = \frac{A_{1,0} - A_{1,1}}{\sqrt{2}}, \quad \tilde{A}_{1,1} = \frac{A_{1,0} + A_{1,1}}{\sqrt{2}}. \quad (8)$$

Here A_{ij} for $i = 1, 2, \dots, N$ and $j = 0, 1, 2$ are the observables measured by the external Alices.

Let us now briefly discuss the main properties of these Bell inequalities. First, it is direct to observe that their maximal values attainable using local deterministic strategies β_C , often referred to as the classical bound, are the same for any l and amount to $\beta_C = (\sqrt{2} + 1)(N-1)$ (Fact 1 in the Supplementary Information provides the proof). Second, the maximal quantum values of I_l , referred to as the quantum or Tsirelson's bounds, equal $3(N-1)$ and are achieved by the following observables of the external parties as

$$A_{1,0} = \frac{x+z}{\sqrt{2}}, \quad A_{1,1} = \frac{x-z}{\sqrt{2}}, \quad A_{1,2} = Y, \quad (9)$$

$$A_{i,0} = Z, \quad A_{i,1} = X, \quad A_{i,2} = Y$$

with $i = 2, \dots, N$ and the Greenberger–Horne–Zeilinger (GHZ)-like states

$$|\phi_l\rangle = \frac{1}{\sqrt{2}} (|l_1 \dots l_N\rangle + (-1)^{l_1} |\bar{l}_1 \dots \bar{l}_N\rangle), \quad (10)$$

where $l_1, l_2, \dots, l_N = 0, 1$ and $\bar{l}_i = 1 - l_i$ (Fact 2 in the Supplementary Information provides the proof). Notice that the measurements in equation (9) form a two-dimensional tomographically complete set of measurements for every party. We now comprise the result in the following theorem:

Theorem 1. Consider the scenario depicted in Fig. 1. If the Bell inequalities I_l (equation (7)) for any l are maximally violated when Eve chooses the input $e = 0$ with each outcome occurring with probability $\bar{P}(l|0) = 1/2^N$, then the measurements of the external parties are equivalent, according to equation (5), to the reference measurements in equation (9), whereas the states $|\psi_{A_i E_i}\rangle$ are equivalent in the sense of equation (4) to the two-qubit maximally entangled state $|\phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$.

Sketch of the proof

Here we provide a sketch of the proof (the full version is provided in Appendix A of the Supplementary Information). First, Eve performs the measurement corresponding to the input $e = 0$ and observes an outcome l , in which way she prepares a post-measurement state shared by the external parties. Now, the observation of the maximal violation of the Bell inequalities I_l by the external parties allows one to self-test the post-measurement state as well as the external parties' measurements. Then, the additional condition that the probabilities of obtaining the outcome l by Eve when measuring $E_0 = \{R_{l|0}\}$ is $1/2^N$ enables self-testing the states generated by the sources. This, in turn, also allows us to certify that Eve's measurement E_0 is the GHZ basis $\{|\phi_l\rangle\langle\phi_l|\}$, where $|\phi_l\rangle$ are given in equation (10).

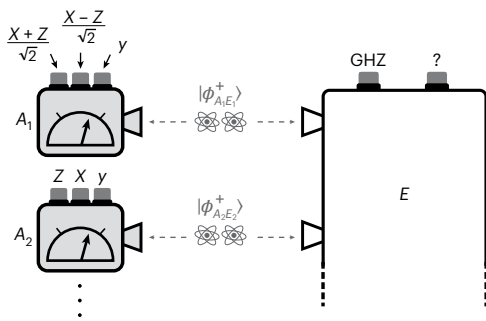


Fig. 2 | Self-testing any quantum measurement. First, we have self-tested all the measurements performed by each external party A_i as equation (9), as well as the sources as the maximally entangled two-qubit state $|\phi_{A_i E_i}^+\rangle$. Thus, Eve's first measurement is certified to the one in the GHZ basis (equation (10)). These are the ingredients sufficient to certify Eve's arbitrary second measurement.

Part 2

Now that the measurements performed by the external parties as well as the states generated by the sources are self-tested, we can show how the quantum network can be used to self-test any extremal POVM in the second measurement E_1 performed by the central party (Fig. 2). Recall that a measurement is extremal if it cannot be expressed as a convex combination of other POVMs (ref. 32 provides a precise definition and a couple of criteria allowing one to decide whether a given measurement is extremal).

For this purpose, let us assume that Eve's reference measurement $E_1 = \{R_{i1}^l\}$ is now some K -outcome extremal POVM defined on $\mathcal{H}_N = (\mathbb{C}^2)^{\otimes N}$. Now, to self-test this reference measurement in E_1 , the parties check whether for all outcomes $l = 0 \dots K-1$ and sequences $i_1 \dots i_N$ with $i_j = 0, 1, 2, 3$, the correlations $\vec{p}$ satisfy the following conditions:

$$\left\langle \tilde{A}_{i_1 i_1} \otimes \bigotimes_{k=2}^N A_{k i_k} \otimes (R_{l1})_E \right\rangle = f_{i_1 \dots i_N}^l, \quad (11)$$

where $f_{i_1 \dots i_N}^l$ are real numbers coming from the decompositions of the measurement operators R_{i1}^l in the N -fold tensor products of the qubit Pauli matrices (Appendix B of the Supplementary Information). We can, thus, state our next result.

Theorem 2. Consider the scenario depicted in Fig. 2, with states generated by the sources and measurements of the external parties being self-tested as in Theorem 1. If the observed correlations $\vec{p}$ additionally satisfy the conditions in equation (11) for Eve's measurement $E_1 = \{R_{i1}^l\}$, then this measurement is self-tested to be the reference extremal POVM E_1 , according to the definition in equation (5).

The proof of this statement is provided in Appendix B of the Supplementary Information. Importantly, the above statement can be extended to any extremal measurement defined on a D -dimensional Hilbert space with arbitrary D . In fact, one can always embed the measurement $\{R_{i1}^l\}$ into $\mathcal{H}_N = (\mathbb{C}^2)^{\otimes N}$, where N is the smallest natural number such that $D \leq 2^N$. Then, to make the measurement fully supported on $\mathcal{H}_N$, one completes it with an additional projector $M^\perp$ such that $\sum_i R_{i1}^l + M^\perp = \mathbb{1}_{2^N}$. This gives rise to a $(K+1)$ -outcome extremal measurement to which Theorem 2 applies. Note that in the case of composite measurements acting on $\bigotimes_i \mathbb{C}^{D_i}$, each of the i th local Hilbert spaces need to be embedded in $(\mathbb{C}^2)^{\otimes n_i}$ such that $\sum_i n_i = N$.

We, thus, obtain a general way of certifying any extremal generalized measurement in the quantum networks scenario. Our scheme is based on the violation of a class of simple Bell inequalities (equation (7)) and some additional conditions (equation (11)) that the observed correlations $\vec{p}$ must satisfy. It is, however, worth noticing that the generality of our scheme comes at the price of the complexity that

grows fast with the dimension of the measurement under consideration and, thus, with the number of involved external parties.

Indirect way to self-test non-extremal measurements

Any non-extremal POVM E can be expressed as a convex combination of extremal POVMs E_e^{ex} , that is, $E = \sum_e p_e E_e^{\text{ex}}$ (ref. 32). Consequently, a way to implement this non-extremal POVM in the star network in Fig. 1 is to let Eve have s inputs $e = 1 \dots s$ corresponding to the measurements E_e^{ex} , each chosen with probability p_e . Now, the above self-testing scheme for extremal measurements (Theorem 2) can be used to self-test any of Eve's measurements corresponding to inputs $e = 1 \dots s$ according to the definition in equation (5). Consequently, one has an indirect way to self-test any non-extremal POVM in the star network.

Part 3

Finally, we are going to show now that by using the above setup, one can certify any (even mixed) quantum state. Consider again the network scenario depicted in Fig. 2. It is clear that Eve could remotely prepare different quantum states with the external parties by performing the concerned quantum measurement on the maximally entangled states generated by the sources. Thus, without loss of generality, the sources along with Eve can be considered a new preparation device. For instance, consider that Eve performs a projective measurement $E_1 = \{|e_1\rangle\langle e_1| \dots |e_K\rangle\langle e_K|\}$. Then, the post-measurement state shared by the external parties when she obtains an outcome l is $|e_l^*\rangle\langle e_l^*|$, where $*$ represents the complex conjugate. As a corollary to the previous statements, we can state the following.

Corollary 1. Consider the scenario depicted in Fig. 2. If the states generated by the sources along with the measurements performed by the external parties and Eve maximally violate the Bell inequalities (equation (7)) and satisfy the statistics (equation (11)) when Eve's reference measurements are extremal, then the post-measurement states with the external parties can be self-tested according to equation (6).

The proof of this statement is provided in Appendix C of the Supplementary Information. As described below equation (6), the state only occurs with a probability $p(l|e=1)$, and thus, the success probability of self-testing the concerned state is $p(l|e=1)$. Let us remark that all Alices and Eve need to communicate at the end of the protocol to obtain the joint probability distribution $\vec{p}$. Consequently, the outcomes of Eve are known for each round after the experiment, and thus, the corresponding statistics allow for DI-certified remote preparation without any additional information from Eve. It is important to emphasize here that our approach of certified remote state preparation also allows the DI certification of separable states and mixed states of any local dimension with the external parties, according to equation (6).

It is simple to observe from the above corollary that any pure state $|\psi\rangle$ can be self-tested with our scheme by appropriately choosing Eve's projective measurement such that one of its measurement elements, say R_{01}^l , is exactly $R_{01}^l = |\psi\rangle\langle\psi|$.

Importantly, however, the scheme is not restricted to pure states only and can be straightforwardly extended to mixed states. Consider a reference mixed state $\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|$ acting on $\mathbb{C}^d$, which we aim to self-test. For this purpose, Eve needs to perform an extremal $3d$ -outcome POVM in the Hilbert space $\mathbb{C}^{2d}$ on her share of the joint state $|\psi_{AE}\rangle$ to create ρ at the external parties' laboratories with the aid of post-processing. To construct this POVM, we first define two sets of d mutually orthogonal vectors $\{|\psi_k\rangle\}$ and $\{|\phi_k\rangle\}$ such that $\langle\psi_k|\phi_{k'}\rangle = 0$ for any k, k' . One then considers a pair $\{|\psi_k\rangle, |\phi_k\rangle\}$ to construct a three-outcome trine POVM²³ as

$$\begin{aligned} M_{k,1} &= p_k |\psi_k\rangle\langle\psi_k|, \quad M_{k,2} = \alpha_k |\tau_{k,2}\rangle\langle\tau_{k,2}|, \\ M_{k,3} &= \beta_k |\tau_{k,3}\rangle\langle\tau_{k,3}|, \end{aligned} \quad (12)$$

where for all k , the vectors $|\tau_{k,2}\rangle$ and $|\tau_{k,3}\rangle$ are certain linear combinations of the vectors $|\psi_k\rangle$ and $|\phi_k\rangle$ whose explicit forms are given in Supplementary Equation (164) and $\alpha_k = \beta_k = (2 - p_k)/2$. The desired extremal $3d$ -outcome rank-one POVM is the one that is composed of all the trine POVM's, that is, $E'_1 = \{M_{k,m}\}_{k,m}$.

Now, when Eve performs this measurement on her share of the joint state and obtains the outcomes $l \equiv (k, 1)$ with probability $p_k/2^N$, the post-measurement (normalized) states of the external parties are given by $\tilde{\rho}_{(k,1)} = |\psi_k\rangle\langle\psi_k|$. These states are self-testable using Corollary 1. Then, the average state when Eve obtains the outcomes $(k, 1)$ over all k is, thus, certified to be

$$\begin{aligned} U_A \rho_A U_A^\dagger &= \frac{1}{\sum_k p(k,1)} \sum_k p(k,1) U_A \rho_A^{(k,1)} U_A^\dagger \\ &= \left(\sum_k M_{(k,1)} \right) \otimes \tilde{\rho}_{A''} = \rho_{A'} \otimes \tilde{\rho}_{A''}, \end{aligned} \quad (13)$$

or its complex conjugate, which is exactly what we promised. Let us say that the external parties want to perform a task with the state ρ . Then, they perform their task with all the post-measurement states of Eve and after the task is done, just like in Bell's scenario where Eve announces the outcomes. Now, all the parties only consider the statistics when Eve obtains $(k, 1)$, and discard the rest of the statistics.

In practice, it is never possible to obtain the exact statistics required for the self-testing of states and measurements. In Appendix D of the Supplementary Information, we analyse the effect of noise on the self-testing correlations. In particular, we show in Theorem 4 of the Supplementary Information that if Eve's actual measurement corresponding to $e = 1$ in the device deviates from the ideal one, with some error ε , then the observed correlations in equation (11) will be close to the ideal ones with an error that is a function of $\sqrt{\varepsilon}$ and the number of external parties N . For our analysis, we do not impose any restrictions on the states generated by the source, the measurements of the external parties or the noise model. Moreover, the technique is simple and, thus, can be adapted to various other self-testing schemes. For Eve's measurement with input $e = 0$, the proof in ref. 20 for $N = 3$ can be straightforwardly adapted to an arbitrary N , allowing one to show that if the observed values of the functionals $\mathcal{J}_l$ are ε -close to the optimal values, then the measurement is $\sqrt{2\varepsilon}$ close to the ideal GHZ-state measurements.

Discussion

After completing our work, we have identified several interesting follow-up questions. First of all, our certification scheme assumes that the sources are independent, and therefore, one crucial follow-up question is whether the scheme can be generalized to the case in which the sources are not independent and exhibit classical or quantum correlations. A possible approach is to give Eve additional inputs, which could then be used to self-test each source, ensuring it produces a maximally entangled state up to some ancillary junk state. Thus, the sources would be independent on the support of the maximally entangled states. However, the states could still be entangled through the junk part and, thus, would require more involved techniques beyond the scope of this work. Additionally, our current scheme requires highly entangled states and measurements, which can be costly to generate. Therefore, it would be interesting to explore the possibility of self-testing quantum states and measurements using partially entangled states and measurements. A possible way is to generalize the idea of self-testing the GHZ basis stated above to the other measurements is through finding Bell inequalities that are maximally violated by all post-measured states corresponding to the measurement elements in the other basis. A simpler problem in this regard will be to consider partially entangled states in the network and then find Bell inequalities that are maximally violated by the corresponding post-measured states, which will again be some partially entangled states. The simplest star network, which is composed of two external parties and a central

party, has been implemented in refs. 28–30 on optical platforms. Our scheme requires one to extend this setup to the scenario with a larger number of external parties. However, we recognize that performing a GHZ-basis measurement is challenging to implement in an optical setup. Thus, it will be highly relevant to provide alternate schemes that do not require GHZ measurements.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41567-026-03181-y>.

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Author contributions

S.S. and R.A. conceived the idea. S.S., A.C.O.J. and R.A. designed the proofs and prepared the paper.

Competing interests

The authors declare no competing interests.

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